Benjamin Choon Heng Lee | Curriculum Vitæ

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Research Interests

I am a Senior Associate Research Scientist as part of the Global Technology Applied Research XR team at JPMorgan Chase. I conduct research in VR/AR to better understand how the technology can best be used across the firm, whether it be by employees in the workplace or by external clients and customers. I have a particular interest in designing novel ways of using VR/AR to allow people to visualize, interact with, and understand data – also known as immersive analytics.

Research Experience

JPMorgan Chase, New York, NY, USA — June 2024 - Present

Senior Associate Research Scientist

- Working with internal stakeholders across the firm's lines of business to address existing problems and/or identify new opportunities in leveraging AR/VR to enhance employee and/or customer experience
- Educating and demonstrating practical use cases of AR/VR at firmwide events and conferences and to senior leadership and executives
- Publishing at academic venues including IEEE VIS [under review] and ACM CHI [c2] and IMWUT [under review]

University of Stuttgart, Stuttgart, Germany — Feb 2023 - Apr 2024

Postdoctoral Researcher

- Conducted research in human-computer interaction, immersive & situated analytics, hybrid user interfaces, and data-driven storytelling
- Co-supervised and mentored PhD students (ongoing as external)
- Shaped and guided projects from idea to publication at venues including IEEE VIS [c7] and ACM CHI [c1, c6]

Microsoft Research, Redmond, WA, USA — Jun 2019 – Sep 2019

Research Intern

- Conducted research in envisioning novel ways of communicating data to audiences through visceral experiences in VR [c13]
- Mentors: Dave Brown, Steven Drucker, Bongshin Lee

Monash University, Melbourne, Australia — Feb 2019 – Jan 2023

Ph.D. in Immersive Analytics

- Conducted research in the combined use of flat surfaces with the virtual 3D space enabled by immersive head-mounted devices for the purposes of interactive data visualisation and immersive analytics [c8, c9, c14]

Education

Monash University, Melbourne, Australia

Feb 2019 – Jan 2023

Ph.D. in Immersive Analytics

- Thesis title: Surfaces and Spaces in Immersive Analytics
- Advisors: Prof. Tim Dwyer, A/Prof. Bernhard Jenny, Dr. Maxime Cordeil, Dr. Arnaud Prouzeau
- Examiners: Prof. Harald Reiterer, Prof. Luciana Nedel

Monash University, Melbourne, Australia

Feb 2015 – Nov 2018

Bachelor of Informatics and Computation Advanced (First Class Honours)

- Thesis title: Heterogeneous Mixed-Reality Display Environments for Immersive Visual Analytics
- Advisors: Prof. Tim Dwyer, A/Prof. Bernhard Jenny, Dr. Maxime Cordeil

Teaching

Units / Courses

S2 ‘21 – S1 ‘22 **FIT5147 Data Visualisation and Exploration (Head TA)**
- Monash University, Australia

S1 2021 **FIT5147 Data Visualisation and Exploration**
- Monash University, Australia

S2 2020 **FIT3146 Maker Lab**
- Monash University, Australia

S1 2020 **FIT5147 Data Visualisation and Exploration**
- Monash University, Australia

Guest Lectures / Invited Talks

Spring 2025 **Web-based Immersive Analytics with Anu.js**
- New York University, NY, USA

Fall 2024 **Immersive Analytics: Data Visualization and Exploration using Immersive Technologies**
- New York University, NY, USA

June 2023 **Immersive and Situated Analytics**
- University of Stuttgart, Germany

May 2023 **Immersive and Situated Analytics**
- Graz University of Technology, Austria

Supervision

PhD Students

Since 2024 **Co-advisor of Tobias Rau**
- With Prof. Michael Sedlmair
- Topic: AR for computational chemistry

Since 2023 **Co-advisor of Carlos Quijano-Chavez**
- With Prof. Dieter Schmalstieg and Prof. Michael Sedlmair
- Topic: Interaction techniques for situated analytics

Since 2023 **Co-advisor of Nina Doerr**
- With Prof. Michael Sedlmair and Prof. Dieter Schmalstieg
- Topic: Visual highlighting in the real world

2023 - 2025 **Co-advisor of Dr. Xingyao Yu**
- With Prof. Michael Sedlmair
- Thesis title: On-Body Visualization with Extended Reality: Exploration and Application

Master's Students

2019 **Co-advisor of Xiaoyun Hu**
- Thesis title: Collaborative Data Visualisation in Virtual Reality

Bachelor's Students

2023 - 2024 **Advisor of Vivien Schraitle**
- Thesis title: Supporting Cross-reality Transfer of Virtual Objects in Hybrid Desktop and Augmented Reality Environments

Academic Service

Reviewing for Conferences (Full Papers)

2025	CHI, ISMAR, VIS, VR
2024	CHI, EuroVis, ISMAR, ISS, MobileHCI, SIGGRAPH, VIS, VR, VRST
2023	ISMAR, ISS, UIST, VIS, VR, VRST
2022	CHI, ISMAR, MobileHCI, VIS, VR
2021	CHI, EuroVis, ISMAR, ISS, VIS
2020	VIS
2019	CHI

Reviewing for Journals

2025	TVCG
2024	IJHCI, TVCG
2023	Frontiers, IJHCI, JCSS, TVCG

Reviewing for Conferences (Short Papers & Workshops)

2025	CHI LBW
2024	alt.CHI, CHI LBW, PacificVis VisNotes, xrWORKS
2023	HybridUI

Organisation & Volunteering

2025	VIS Full Paper Program Committee
2023	Co-organiser of HybridUI workshop at ISMAR
2022	SV at VR (online)
2020	SV at OzCHI (online)

Awards and Recognition

2025	ACM CHI Best Paper Award [c1]
2023	Monash University Vice-Chancellor's Commendation For Thesis Excellence Award
2022	ACM CHI Honourable Mention Award [c9]
2021	ACM ISS Honourable Mention Award [c12] IEEE VIS Honourable Mention Award [c13]

Conference and Journal Papers

** denotes equal contribution*

c1. Tobias Rau, Tobias Isenberg, Andreas Köhn, Michael Sedlmair, and **Benjamin Lee**. 2025. Traversing Dual Realities: Investigating Techniques for Transitioning 3D Objects between Desktop and Augmented Reality Environments. *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI 2025). https://doi.org/10.1145/3706598.3713949. **Best Paper Award (top 1% of submissions).**

c2. Matt Gottsacker, Mengyu Chen, David Saffo, Feiyu Lu, **Benjamin Lee**, and Blair MacIntyre. 2025. Examining the Effects of Immersive and Non-Immersive Presenter Modalities on Engagement and Social Interaction in Co-located Augmented Presentations. *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI 2025). https://doi.org/10.1145/3706598.3713346.

c3. Xingyao Yu, David Rosin, Johannes Kässinger, **Benjamin Lee**, Frank Dürr, Christian Becker, Oliver Röhrle, and Michael Sedlmair. 2024. PerSiVal: On-Body AR Visualization of Biomechanical Arm Simulations. *IEEE Computer Graphics and Applications.* https://doi.org/10.1109/MCG.2024.3494598.

c4. Xiaoyan Zhou*, **Benjamin Lee***, Francisco R. Ortega, Anil Ufuk Batmaz, and Yalong Yang. 2024. Lights, Headset, Tablet, Action: Exploring the Use of Hybrid User Interfaces for Immersive Situated Analytics. *Proceedings of the ACM on Human-Computer Interaction* (ISS 2024). https://doi.org/10.1145/3698147.

c5. Nina Doerr, **Benjamin Lee**, Katarina Baricova, Dieter Schmalstieg, and Michael Sedlmair. 2024. Visual Highlighting for Situated Brushing and Linking. *Computer Graphics Forum* (EuroVis 2024). https://doi.org/10.1111/cgf.15105.

c6. Xingyao Yu, **Benjamin Lee**, and Michael Sedlmair. 2024. Design Space of Visual Feedforward and Corrective Feedback in XR-Based Motion Guidance Systems. *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (CHI 2024). https://doi.org/10.1145/3613904.3642143.

c7. **Benjamin Lee,** Michael Sedlmair, and Dieter Schmalstieg. 2023. Design Patterns for Situated Visualization in Augmented Reality. *IEEE Transactions on Visualization and Computer Graphics* (VIS 2023). https://doi.org/10.1109/TVCG.2023.3327398.

c8. **Benjamin Lee**, Arvind Satyanarayan, Maxime Cordeil, Arnaud Prouzeau, Bernhard Jenny, and Tim Dwyer. 2023. Deimos: A Grammar of Dynamic Embodied Immersive Visualisation Morphs and Transitions. *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (CHI 2023). https://doi.org/10.1145/3544548.3580754.

c9. **Benjamin Lee**, Maxime Cordeil, Arnaud Prouzeau, Bernhard Jenny, and Tim Dwyer. 2022. A Design Space For Data Visualisation Transformations Between 2D And 3D In Mixed-Reality Environments. *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (CHI 2022). https://doi.org/10.1145/3491102.3501859. **Honourable Mention Award (top 5% of submissions).**

c10. Kadek Ananta Satriadi, Jim Smiley, Barrett Ens, Maxime Cordeil, Tobias Czauderna, **Benjamin Lee**, Ying Yang, Tim Dwyer, and Bernhard Jenny. 2022. Tangible Globes for Data Visualisation in Augmented Reality. *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (CHI 2022). https://doi.org/10.1145/3491102.3517715.

c11. Ying Yang, Tim Dwyer, Michael Wybrow, **Benjamin Lee**, Maxime Cordeil, Mark Billinghurst, and Bruce H. Thomas. 2022. Towards Immersive Collaborative Sensemaking. *Proceedings of the ACM on Human-Computer Interaction* (ISS 2022). https://doi.org/10.1145/3567741.

c12. Jim Smiley, **Benjamin Lee**, Siddhant Tandon, Maxime Cordeil, Lonni Besançon, Jarrod Knibbe, Bernhard Jenny, and Tim Dwyer. 2021. The MADE-Axis: A Modular Actuated Device to Embody the Axis of a Data Dimension. *Proceedings of the ACM on Human-Computer Interaction* (ISS 2021). https://doi.org/10.1145/3488546. **Honourable Mention Award (top 5% of submissions).**

c13. **Benjamin Lee**, Dave Brown, Bongshin Lee, Christophe Hurter, Steven Drucker, and Tim Dwyer. 2021. Data Visceralization: Enabling Deeper Understanding of Data Using Virtual Reality. *IEEE Transactions on Visualization and Computer Graphics* (VIS 2021). https://doi.org/10.1109/TVCG.2020.3030435. **Honourable Mention Award (top 5% of submissions).**

c14. **Benjamin Lee**, Xiaoyun Hu, Maxime Cordeil, Arnaud Prouzeau, Bernhard Jenny, and Tim Dwyer. 2021. Shared Surfaces and Spaces: Collaborative Data Visualisation in a Co-Located Immersive Environment. *IEEE Transactions on Visualization and Computer Graphics* (VIS 2021). https://doi.org/10.1109/TVCG.2020.3030450.

Short Papers (Extended Abstracts, Posters, Workshops, and Demos)

s1. Ying Yang, Tim Dwyer, Zachari Swiecki, **Benjamin Lee**, Michael Wybrow, Maxime Cordeil, Teresa Wulandari, Bruce H. Thomas, and Mark Billinghurst. 2024. Putting Our Minds Together: Iterative Exploration for Collaborative Mind Mapping. *Proceedings of the Augmented Humans International Conference 2024* (Posters). https://doi.org/10.1145/3652920.3653043.

s2. Carlos Quijano-Chavez, Nina Doerr, **Benjamin Lee**, Dieter Schmalstieg, and Michael Sedlmair. 2024. Brushing and Linking for Situated Analytics. *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*. https://doi.org/10.1109/VRW62533.2024.00116.

s3. Xiaoyan Zhou, Yalong Yang, Francisco Ortega, Anil Ufuk Batmaz, and **Benjamin Lee**. 2023. Data-driven Storytelling in Hybrid Immersive Display Environments. *2023 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. https://doi.org/10.1109/ISMAR-Adjunct60411.2023.00056.

s4. Anika Sayara, **Benjamin Lee**, Carlos Quijano-Chavez, and Michael Sedlmair. 2023. Designing Situated Dashboards: Challenges and Opportunities. *2023 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. https://doi.org/10.1109/ISMAR-Adjunct60411.2023.00028.

s5. Ari Kouts, Lonni Besançon, Michael Sedlmair, and **Benjamin Lee**. 2023. LSDvis: Hallucinatory Data Visualisations in Real World Environments. *alt.VIS Workshop* (VIS 2023). https://doi.org/10.48550/arXiv.2312.11144.

s6. Nicholas Spyrison, **Benjamin Lee**, and Lonni Besançon. 2021. “Is IEEE VIS *that* Good?” On Key Factors in the Initial Assessment of Manuscript and Venue Quality. *alt.VIS Workshop* (VIS 2021). https://doi.org/10.31219/osf.io/65wm7.

s7. **Benjamin Lee**, Maxime Cordeil, Arnaud Prouzeau, and Tim Dwyer. 2019. FIESTA: A Free Roaming Collaborative Immersive Analytics System. *Proceedings of the 2019 ACM International Conference on Interactive Surfaces and Spaces* (ISS 2019). https://doi.org/10.1145/3343055.3360746.